

Other Numerical Simulation & Modelling

070 Bhadra

7. What are the properties of Random number? Using Chi-Square test the uniformity at 90% for the given random numbers. Degree of freedom for 6 = 10.645, 7 = 12.017, 8 = 13.362, 9 = 14.684, 10 = 15.987.

[3+6]

20	34	43	42	14	10	33	17	6	11
15	16	4	1	35	22	9	46	37	57
51	49	40	27	59	5	44	19	41	55
53	29	3	31	48	8	56	28	12	7

074 Magh

7. Write an algorithm for Gap Test. A sequence of 10,000 five digit numbers has been generated and analysis indicates the following combinations and frequencies. Based on Poker Test check whether the number are independent. Use $\chi^2_{0.05,6} = 12.592$

[4+6]

Combinations	Observed Frequencies
All Different	3044
One pair	5020
Two pair	1090
Three of a kind	700
Full house	95
Four of a kind	40
Five of a kind	11
Total	10,000

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6. A sequence of 1000 four digit number has been generated and an analysis indicates the following combination and frequencies:

[8]

Combination (i)	Observed Frequency (O _i)
Four different digit	520
One pair	390
Two pair	55
Three like digit	34
Four like digit	1
	1000

072 Ashwin

7. A sequence of 10,000 five digital numbers has been generated and analysis indicates the following combinations and frequencies. [8]

Combinations (i)	Observed Frequency (O_i)
All different	3450
One pair	4510
Two pair	1150
Three of a kind	750
Full house	85
Four of a kind	40
Five of a kind	15
Total	10,000

Based on Poker Test Check whether the number are independent. Use $\alpha = 0.05$ and $N = 6$ is 12.592

072 Magh

7. A sequence of 1000 four digit numbers has been generated and an analysis indicates the following conditions. Based on poker test, are these numbers independent? (use $\alpha=0.05$ and $N = 4$ is 9.488) [8]

Combination Distributions	Observed Frequency
4 different Digits	510
3 Like Digits	40
4 Link Digits	5
1 Pair	410
2 Pair	35

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7. Define pseudo random numbers. The following numbers have been generated 0.44, 0.19, 0.88, 0.27, 0.55, 0.13, 0.63, 0.74, 0.11 and 0.33. Use the Kolmogorov-Smirnov test with $\alpha = 0.05$ to determine, if the hypothesis that the numbers are uniformly distributed on the interval $[0, 1]$ can be rejected. (Note that the critical value of D for $\alpha 0.05$ and $N = 10$ is 0.410. [2+6]

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- b) A set of 10,000 4-digit random values have been generated. An observation shows than 5065 values have all different digits, 2000 have 2 of a kind digits, 760 have 3 of a kind, 1500 have 2 pairs and 675 have all same digits. Test these values for randomness using Poker test (Use $\alpha = 0.05$). [7]

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- b) A set of 100 random numbers have been generated as below:

[2+3]
[6]

0.23 0.30 0.07 0.39 0.43 0.47 0.66 0.19 0.49 0.37 0.53
0.80 0.56 0.10 0.28 1.00 0.23 0.04 0.93 0.25 0.90 0.24
0.88 0.75 0.54 0.84 0.42 0.68 0.12 0.68 0.16 0.72 0.93
0.14 0.71 0.23 0.76 0.96 0.89 0.82 0.99 0.42 0.22 0.77
0.36 0.95 0.35 0.14 0.89 0.26 0.99 0.78 0.23 0.75 0.67
0.34 0.04 0.95 0.41 0.89 0.64 0.88 0.92 0.62 0.84 0.36
0.88 0.32 0.86 0.24 0.42 0.54 0.35 0.60 0.88 0.84 0.15
0.86 0.99 0.63 0.56 0.16 0.53 0.86 0.69 0.74 0.70 0.38
0.48 0.67 0.54 0.97 0.50 0.09 0.20 0.72 0.27 0.72 0.96

Test these values for uniform distribution using Chi Square Test (Use $\alpha=0.05$)

c) A sequence of 1000 four digit numbers has been generated and analysis indicates the following combinations and frequencies:

Combination (i)	Observed Frequency (O_i)
Four different digits	540
One pair	380
Two pairs	54
Three like digits	25
Four like digits	1
	1000

[7]

Based on poker test check whether the numbers are independent. Use $\alpha=0.05$ and $\chi^2 = 9.49$

7. Explain the iterative process of calibration of model.

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6. A sequence of 1000 four digit numbers has been generated and analysis indicates the following combinations with frequencies:

Combination (i)	Observed Frequency (O_i)
Four different digits	570
One pair	380
Two pairs	34
Three like digits	15
Four like digits	1
	1000

Based on poker test check whether the numbers are independent. Use $\alpha=0.10$ and $N=4$ is 7.78

(8)

